Integrated Retrieval of the Temperature and Humidity Profiles of Atmospheric Boundary Layer by Combining Ground-Based Infrared Hyperspectral Interferometers and Microwave Radiometers

Yao Xiao, Shuai Hu, Wanxia Deng, Ruijun Dang, Lei Liu, Wei Huang, and Wanying Yang

***Abstract*— Atmospheric temperature and humidity profiles are the basic parameters used to describe the vertical distribution of atmospheric states. Continuous observations of accurate temperature and humidity profiles are essential for exploring boundary layer thermal and dynamic characteristics. To this end, an intelligent retrieval algorithm (IReA) based on a convolutional neural network (CNN) is proposed to retrieve atmospheric temperature and humidity profiles by combining observations from ground-based infrared hyperspectral radiometers and microwave radiometers (MWRs). The results show that the inclusion of microwave observations can effectively improve the retrieval accuracy of temperature and humidity profiles relative to the results from atmospheric emitted radiance interferometer (AERI) under clear-sky conditions, where the root mean square error (RMSE) of the temperature profile is 0.79 K and the RMSE of the humidity profile is 0.95 g/kg. The accuracies of different retrieval methods are also evaluated. In general, the RMSE derived from IReA is improved by at least 9% compared to the results from the physical retrieval method and BP neural network method. Given that clouds are semitransparent in the microwave region, the retrieval accuracy of the temperature and humidity profile of IReA are also improved under cloudy conditions when microwave observations are introduced.**

***Index Terms*— Atmospheric emitted radiance interferometer (AERI), convolutional neural network (CNN), humidity, microwave radiometer (MWR), temperature.**

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I. Introduction

THE atmospheric boundary layer is not only an important transport channel for atmospheric materials, momentum, and energy but also the main layer for the emission, diffusion, and transformation of atmospheric pollutants [1]. Due to the effect of atmospheric turbulence, the daily variation in meteorological elements such as temperature and humidity is significant, with time scales ranging from 1 h to 1 day [2], [3], [4]. Therefore, to accurately study the thermal and dynamic characteristics of the boundary layer, obtaining temperature and humidity profiles with high temporal resolution has become the basic data requirement for boundary layer research.

At present, the atmospheric temperature and humidity profiles of the boundary layer are mainly obtained by radiosonde and ground-based remote sensing methods [5], [6]. Radiosondes can directly measure atmospheric profiles, but their temporal time resolution is low, and single detection is expensive. In contrast, ground-based remote sensing has unique advantages because of its continuous observation and high-cost performance. In particular, with the development of the Fourier transform infrared hyperspectral spectroradiometer, thousands of spectral channels can be synchronously measured, and the vertical profiles of atmospheric temperature and humidity can also be retrieved accurately with a high temporal resolution [7]. At present, the atmospheric emitted radiance interferometer (AERI) developed by the University of Wisconsin [8] can measure 3–18 μm radiance with a resolution of less than 1 cm^{-1} and has been widely deployed in the atmospheric radiation measurement (ARM) research program [9]. However, it should be noted that, limited by the principle of infrared remote sensing, this method is mainly applicable to inversion under clear-sky conditions [10]. Microwave radiometer (MWR) is also an important means to obtain atmospheric profiles; although its remote sensing channels are much fewer than those of AERI, it also contains abundant atmospheric temperature and humidity information. More importantly, due to the long wavelength of microwaves, they can penetrate clouds and improve the retrieval of temperature and humidity profiles

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under cloudy conditions [11], [12]. Therefore, considering the advantages of AERI and MWR, this article proposes retrieving temperature and humidity profiles by combining microwave observations and ground-based infrared hyperspectral measurements. The purpose of this work includes the following two parts: on the one hand, the information content can be increased by adding microwave atmospheric radiation information to infrared hyperspectral radiation; thus, this approach might enhance the accuracy of atmospheric profiles in clear skies. On the other hand, by combining the penetration ability of microwaves, the retrieval ability of temperature and humidity profiles under cloudy conditions might also be improved.

To realize the inversion of atmospheric temperature and humidity profiles based on ground-based infrared hyperspectral radiometers and MWRs, designing a retrieval algorithm is a very important step. At present, the physical retrieval algorithm is the main method used for the inversion of atmospheric profiles by AERIs. The "onion peeling" algorithm was first proposed [9], but due to the ill-posed characteristics of the inversion problem, the accuracy of the "onion peeling" algorithm is highly dependent on the initial values. To address the drawbacks of the "peeling onion" method, Turner proposed an AERIoe inversion algorithm in 2013 [13]. This method is a Gauss–Newton iterative method based on the Bayesian optimal estimation principle and can effectively improve the accuracy and stability of the retrieval process. Based on this work, Huang et al. [14] proposed the use of the L-curve method to select the regularization operator of the AERIoe retrieval algorithm, which can improve retrieval efficiency. Since the physical retrieval method involves complex simulations of atmospheric radiative transfer [15], [16], its computational complexity is high, and the retrieval time is long. To improve computational efficiency, the statistical inversion method has also become an important research direction [17]. Early on, the eigenvector method was mainly used to infer the atmospheric profiles [18], [19]. However, as the essence of this method is establishing a multiple linear regression relationship between the radiance and atmospheric profiles, the nonlinear characteristics of the radiative transfer processes are not considered enough; therefore, the retrieval accuracy of this method is not very high [11]. In recent years, retrieval algorithms using artificial neural networks (ANNs) and deep learning methods have been gradually developed [20], [21], [22], [23]. Compared with physical retrieval algorithms, deep learning methods do not require complex simulations of radiative transfer, so the retrieval efficiency of deep learning methods is much greater than that of physical retrieval methods. In addition, due to the excellent feature representation ability and generalizability of deep learning methods, nonlinear relationships between atmospheric temperature and humidity and between infrared and microwave radiance can be explored in detail; therefore, the retrieval accuracy can be effectively improved [24]. Yan et al. [25] developed a batch normalization and robust neural network (BRNN) inversion model for MWRs and reported that the inversion of temperature and humidity near the surface improved. For FY-4A/GIIRS, Cai et al. [26] used an ANN to infer the atmospheric temperature and humidity profiles, and the results showed that good agreement was achieved with the AIRS products. Li et al. [27] used data from the Sentinel-3A satellite to invert the LST using a variety of algorithms and evaluated them on different surface types. It shows the direction for the optimization and improvement of subsequent surface temperature inversion algorithms.

The convolutional neural network (CNN) is a new type of deep learning model developed in recent years [28], [29], [30]. Compared with traditional neural networks, this network not only extracts features at different scales and abstraction levels via multilayer convolutional and pooling operations but also effectively handles noise and defects in the input data. Due to its unique advantages, it can realize nonlinear fitting [31] under the conditions of complex background information and uncertain inference rules, and it is now widely used in the field of machine vision [28]. Using a CNN, Malmgren-Hansen et al. [32] accurately retrieved the atmospheric temperature profile of an infrared atmospheric sounding interferometer (IASI) for the first time and verified the unique advantages of this model. Therefore, under the combination of ground-based infrared hyperspectral and MWRs, CNNs can be used to retrieve atmospheric temperature and humidity profiles, so that both the accuracy and efficiency of the retrieval process can be improved compared with those of traditional physical retrieval methods.

The main structure of this article is as follows. In Section II, the AERI and MWR data are introduced. In Section III, the intelligent retrieval algorithm (IReA) based on the CNN was established. In Section IV, the temperature and humidity profiles of the retrieval algorithm, which include clear-sky and cloudy conditions, are discussed and analyzed. Section V discusses situation in which the proposed method may not perform well. Finally, Section VI provides a summary of our work.

II. Data and Instrument

The Southern Great Plains (SGPs) site is the first in situ measurement site in the ARM program. The SGP site is located in the central United States [33], see Fig. 1 for specific locations. There are the largest and most extensive climate research facilities in the world, which is highly important for revealing the characteristics of clouds, aerosols, and the atmosphere [34]. To construct the retrieval model of the combined ground-based infrared hyperspectral and MWR, radiosonde, AERI, and MWR data from the SGP station from 2006 to 2017 were used. In our retrieval experiment, we screened a total of 7351 groups from the years 2011–2016 for training purposes. For validation, we selected 802 groups under clear-sky conditions and 1110 groups under cloud conditions from the year 2017.

The radiosonde was launched from the same location, with the sonde release point located 150 m north of the AERI facility. In general, at the ARM SGP site, radiosondes were launched four times a day, corresponding to the time: 05:30 UTC, 11:30 UTC, 17:30 UTC, and 23:30 UTC [35]. The SGP MWR is a two channel MWR and mainly measures atmospheric microwave radiation at 23.8 and 31.4 GHz [33]. The first channel (23.8 GHz) is located in the wing region

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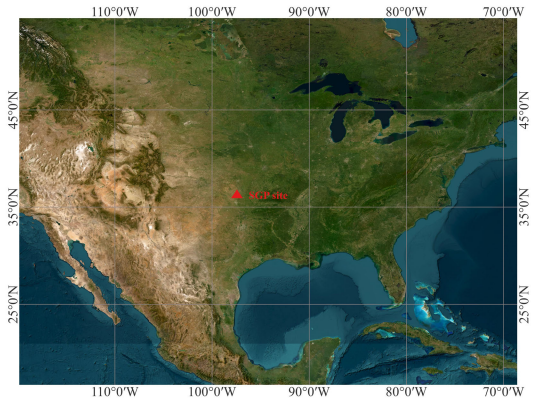


Fig. 1. Specific geographic location of the SGP site.

TABLE I
RETRIEVAL CHANNELS OF AERI ARE USED FOR THE TEMPERATURE AND HUMIDITY PROFILES (cm^{-1})

Water vapor	Temperature
538-588 (104)	612-618 (12)
1250-1350 (207)	624-660 (74)
	674-713 (81)
	2223-2260 (77)

of the 22.2 GHz water vapor absorption band and is mainly used to measure water vapor. The second channel (31.4 GHz) is located in the atmospheric window region and is mainly affected by the liquid water content, and these two frequencies allow simultaneous determination of the water vapor and liquid water content along the selected path with a time resolution of 20 s. The AERI was developed by the Space Science and Engineering Center (SSEC) at the University of Wisconsin-Madison, and AERI can obtain temperature and humidity profiles mainly by receiving atmospheric downwelling infrared radiation, its detection channels range from 520 to 3020 cm^{-1}, and the spectral resolution is about 0.5 cm^{-1}, where the instrument field-of-view is 1.3°. Calibrated sky radiance spectra are produced approximately every 8 min [8]. The retrieval channels of the temperature profiles are mainly distributed near the absorption spectral lines of CO_2, and the humidity retrieval channels generally include water vapor sensitive channels. The specific channels used in this article are shown in Table I [36], where for the retrieval of humidity profiles, a total of 311 channels are used, and for the retrieval of temperature profiles, 244 channels are used.

Because the temporal and vertical resolutions of AERI data are not the same as those of radiosonde data, it is necessary to match the data in temporal and spatial space (the temporal resolution of AERI data is much greater than that of radiosonde data, while the vertical resolution of sonde data is much greater than that of AERI data). Furthermore, similar operations should also be performed on the MWR data. The steps are as follows.

1) *Temporal Matching:* The measurement data obtained from each observation instrument were matched in time, using the time of the sounding release as a benchmark to select the AERI data with the closest time for comparison.
2) *Quality Control:* AERI radiation spectra may contain outliers during the actual measurement, so spectral features with obvious outliers, such as negative radiation and smoothed spectra, were removed for quality control.
3) *Cloud Rejection:* In the retrieval experiment under clear-sky conditions, the presence of clouds can significantly affect the observed AERI radiation, so the laser ceilometer and the AERI radiation data itself were used for cloud detection and selection of clear-sky samples.

In this study, 85% of the temperature profile information and 70% of the water vapor profile information in the observed data are concentrated below 2 km [13], which makes the height of the AERI inversion generally not more than 3 km [4]; therefore, our retrieval altitude is under 3 km.

III. METHODS

To retrieve temperature and humidity profiles by combining AERI and MWR, a retrieval algorithm based on the intelligent retrieval architecture (IReA, which is an intelligent machine learning model) is proposed in this article. The design of the IReA is inspired by the excellent performance of deep CNNs in feature representation, automatic feature extraction, and generalization learning. The essence of the IReA model is to establish nonlinear mapping relationships between the observed radiance of AERI/MWR and the atmospheric profiles obtained by the radiosondes.

Fig. 2 illustrates the Schematic of the IReA. The IReA model is constructed of three modules: the feature extraction module, the feature concatenation module, and the regression module. The feature extraction module is designed to extract the nonlinear features between the AERI and MWR radiance observations and the atmospheric profiles detected by the radiosonde. This module is established by alternately stacking several 1-D convolutions, batch normalization layers (BN), 1-D max-pooling layers, and rectified linear unit (ReLU) layers. The feature concatenation module is used to combine the extracted AERI and MWR feature vectors and provides the parameters required by the regression module. The regression module consists of a fully connected layer and a ReLU layer and is applied to accomplish the inversion process. In this network, the convolutional layer is the core part of the CNN and is applied to extract features from data samples [37]. There are three key parameters in the convolutional layer, namely the convolutional kernel (filter), step length (stride), and boundary padding (padding), which determine the size of the output feature map of the convolutional layer. The BN is designed to accelerate CNN training by normalizing layer inputs [38]. The ReLU function is applied to solve the problem of gradient disappearance in CNNs, which can accelerate the convergence of the retrieval process [39]. The function of the 1-D max-pooling layer is to perform secondary feature extraction by downsampling the feature map to remove redundant information and simplify the complexity of the network [37]. The fully connected layer maps the feature space calculated from the previous layer (convolution, pooling, etc.) to the label space by rearranging the extracted feature maps

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into feature vectors. Because of the special structure of the IReA, the nonlinear relationship between the temperature and humidity profiles and radiation can be effectively determined.

A. Design of the IReA

In the IReA, the observed radiance of the infrared hyperspectral radiometer AERI is $\mathbf{Y}^a = \{\mathbf{y}_i^a\}_{i=1}^N$, and the observed radiation of the MWR is $\mathbf{Y}^m = \{\mathbf{y}_i^m\}_{i=1}^N$, where $\mathbf{y}_i^a$ and $\mathbf{y}_i^m$ represent the ith sample of the observed radiances, and N is the total number of the samples obtained. The corresponding true values are obtained from the radiosonde and are presented in $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i$ represents the measured temperature and humidity measured by the radiosonde at the same time. In the description of the algorithm, for simplicity, the same is used to define the temperature vector $\mathbf{x}_i^t = [t_1, t_2, \ldots, t_m]^{\mathrm{T}}$ or the humidity vector $\mathbf{x}_i^q = [q_1, q_2, \ldots, q_m]^{\mathrm{T}}$, where$t_j$ and q_j represent the temperature and water vapor mixing ratio at the jth vertical, respectively, m represents the total number of vertical heights, and "T" represents the transpose operation of the matrix.

In the implementation of the IReA, the input AERI of the MWR data is first characterized by the feature extraction module, by which the nonlinear relationship between AERI observations $\mathbf{Y}^a = \{\mathbf{y}_i^a\}_{i=1}^N$ and the MWR observations $\mathbf{Y}^m = \{\mathbf{y}_i^m\}_{i=1}^N$ and the corresponding radiosonde data $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N$ can be obtained. The function of this module can be defined as

$$\mathbf{F}_i^a = E_\alpha\left(\mathbf{y}_i^a\right) \tag{1}$$

$$\mathbf{F}_i^m = G_\theta\left(\mathbf{y}_i^m\right) \tag{2}$$

where $E_\alpha(\mathbf{y}_i^a)$ is the feature extraction function of the AERI feature extraction module, which is composed of a convolution layer, BN, pooling layer, and ReLU layer, and its training parameter is assumed to be α. $G_\theta(\mathbf{y}_i^m)$ is the feature extraction function for the MWR, where θ is the training parameter. Because the MWR data only have two channels, unlike $E_\alpha(\mathbf{y}_i^a)$, an upper sampling layer should be added to $G_\theta(\mathbf{y}_i^m)$ to increase the dimensionality of the data.

Then, we input the generated features $\mathbf{F}_i^\alpha$ and $\mathbf{F}_i^m$ of the AERI and MWR data into the feature concatenation module, written as

$$\mathbf{F}_i = \mathrm{Concat.}\left(\mathbf{F}_i^a, \mathbf{F}_i^m\right) \tag{3}$$

where Concat. represents the feature splicing process; its purpose is to transfer feature vectors $\mathbf{F}_i^\alpha$ and $\mathbf{F}_i^m$ into one vector. $\mathbf{F}_i$. Then $\mathbf{F}_i$ is passed to the regression module R_β (β is the training parameter). By using this module, the retrieval values of atmospheric temperature and humidity profiles can be obtained by

$$\hat{\mathbf{x}}_i = R_\beta(\mathbf{F}_i) \tag{4}$$

where $\hat{\mathbf{x}}_i$ is the retrieval temperature and humidity profile corresponding to the observed radiance vector $\mathbf{y}_i$, and the collection of all retrieved profiles is expressed as $\hat{\mathbf{X}} = \{\hat{\mathbf{x}}_i\}_{i=1}^N$.

In the training process of the IReA, the atmospheric profiles obtained by the radiosondes are taken as the supervisory signals; then, by gradually optimizing the training parameters of the IReA, the output parameters (i.e., the inverted profiles) of the IReA approach the temperature and humidity obtained by the radiosondes (the true values). The objective function for the optimization of the IReA can be defined as

$$O(\alpha, \theta, \beta) = \frac{1}{L}\frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \hat{\mathbf{x}}_i\right\|^2 \tag{5}$$

where $O(\alpha, \theta, \beta)$ is the learning function which is determined by optimizing the training parameters α, θ, and β, and L represents the length of the vector $\mathbf{x}_i$.

In this study, we employ stochastic gradient descent (SGD) to optimize model parameters, thereby enhancing the model's performance on training data. SGD minimizes the loss function, enabling the model to fit the training data more effectively. This iterative optimization algorithm approaches the optimal solution by updating parameters over multiple iterations. In each iteration, the algorithm calculates the gradient of the loss function based ON current parameter values and updates the parameters in the opposite direction of the gradient, progressively reducing the value of the loss function.

B. Network Setting

The input data of the IReA include the radiance observed by the 555 channels of AERI (including 244 temperature sensitive channels and 311 water vapor sensitive channels) [20] and two channels of the MWR. The feature extraction module of the IReA established in this article has four 1-D convolution layers (1D CONVs): Conv1 (with 200 channels), Conv2 (with 400 channels), Conv3 (with 500 channels), and Conv4 (with 550 channels). The four 1D CONVs of AERI have convolution kernel sizes of 5, and the first three 1D CONVs of the MWR have convolution kernel sizes of 1. The fourth 1D CONV has a convolution kernel size of 3. Each convolutional layer is followed by BN, ReLU, and a 1-D maximum pooling layer (1D Max Pool). The size of the largest pooling layer is set as 2, and its stride length is taken as 2. For the MWR data features, upsampling is performed after the ReLU of the first and second convolution layers. For the AERI and MWR, the feature splicing module is first used to concatenate them into a whole vector, and then is substituted into the regression module used to process them. The regression module consists of two fully connected layers (FC): FC1 with 1000 neurons and FC2 with 37 neurons. The FC1 layer is followed by the ReLU and dropout layer, and FC2 is also the output layer. The number of convolutional layers is generally determined according to the complexity of the task, usually from simple to complex network structure, we can start with several convolutional layers, and gradually increase, with each additional layer, the expression ability of the network enhanced, but it may also lead to overfitting. For the convolutional kernel, we generally use 3 by 3, 5 by 5, and 7 by 7, which have been verified by experiments. Although a larger convolutional kernel can capture more local features, it is easy to cause overfitting due to the larger number of parameters and the larger amount of computation. We have experimented and the current architecture is the one with the relatively best results. That's why we've chosen it. Fig. 3

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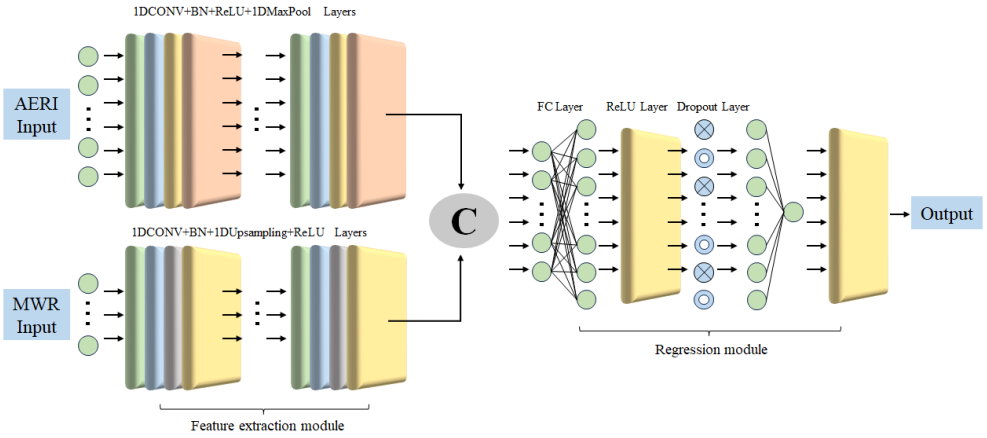


Fig. 2. Schematic of the IReA. 1D CONV: 1-D convolution layer, BN: batch normalization layer, ReLU: activation function layer, 1D Max Pool: 1-D maximum pooling layer, 1D Upsampling: 1-D upper sampling layer, and FC: fully connected layer.

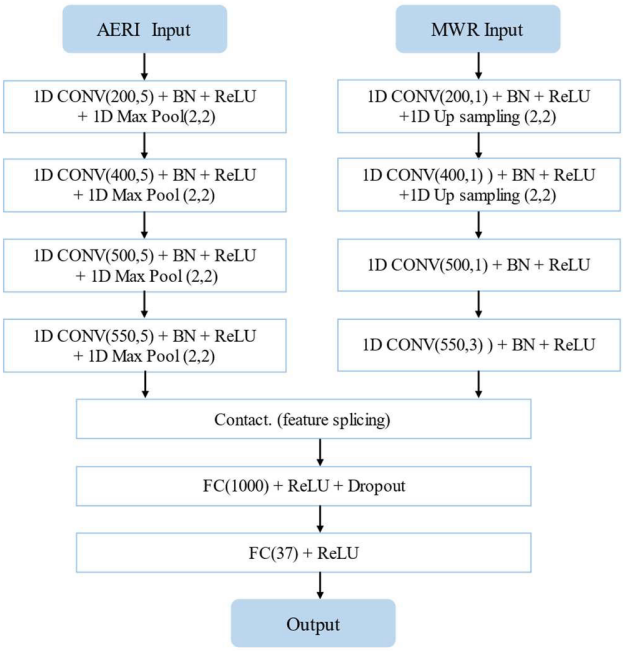


Fig. 3. Detailed illustrations of the IReA for the atmospheric profiles of the retrieval framework.

shows the detailed structure and its parameters. In this article, temperature and water vapor channels represent bands.

C. Evaluation Metrics

To quantitatively analyze the performance of the IReA under clear-sky conditions, we first compare the atmospheric profiles of the three methods through several typical case studies. Subsequently, based on validation datasets (including 802 samples from 2017), the mean bias (BIAS), root mean square error (RMSE), and mean absolute error (MAE) between the results of these three methods and the profiles obtained by the radiosondes are calculated. The three indicators are calculated as follows:

$$\mathrm{BIAS}(i) = \frac{\sum_{j=1}^{M} \left(X_{\mathrm{sonde}}(i,j) - X_{\mathrm{retrieval}}(i,j)\right)}{M} \tag{6}$$

$$\mathrm{RMSE}(i) = \sqrt{\frac{\sum_{j=1}^{M} \left(X_{\mathrm{sonde}}(i,j) - X_{\mathrm{retrieval}}(i,j)\right)^2}{M}} \tag{7}$$

$$\mathrm{MAE} = \frac{1}{M}\sum_{i=1}^{m} \left|X_{\mathrm{retrieval}} - X_{\mathrm{sonde}}\right| \tag{8}$$

where X_{sonde} represents the temperature or humidity profiles obtained by the radiosondes, $X_{\mathrm{retrieval}}$ represents the temperature or humidity retrieved by different methods, j represents the sample sequence of temperature or humidity, i represents the number of altitude layers, and M represents the total number of samples.

D. Other Methods

In this article, we compare the IReA method for the traditional neural network model (BP neural network) and the physical retrieval method under clear-sky conditions, so the BP neural network method and the physical retrieval method are briefly introduced here.

1) BP Neural Networks: The BP method [40] can realize the nonlinear projection from the input observations to the output retrieval profiles. The BP neural network is a multilayer network structure that generalizes the W–H learning rule and trains the weights of nonlinear differentiable functions. In the field of atmospheric science, the BP neural network model has been widely applied. Compared with other neural network models, the BP model has the advantages of good generality, high maturity, and reliability. The BP model consists of two processes: the forward propagation of the signal and the backward propagation of the error.

2) Physical Retrieval Methods: According to Bayesian estimation theory, under the assumption that the observation error and the background error are Gaussian distributed and that they are independent of each other, the retrieval methods used for IR radiation are essentially to find the solution by minimizing the cost function with the following form [41], [42]:

$$J(\mathbf{x}) = \left[\mathbf{y}^m - F(\mathbf{x})\right]^{\mathrm{T}} \mathbf{S}_e^{-1} \left[\mathbf{y}^m - F(\mathbf{x})\right] + \gamma(\mathbf{x} - \mathbf{x}_0)\mathbf{S}_a^{-1}(\mathbf{x} - \mathbf{x}_0) \tag{9}$$

where y^m and $F(X)$ denote the radiation observed by AERI and the radiation value calculated from the radiative transfer

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model, respectively. $\mathbf{x}$ represents the atmospheric profiles to be retrieved, including temperature profiles and humidity profiles. x_0 is the initial profile. **Se** and **Sa** represent the observation error covariance matrix and background error covariance matrix, respectively. γ is a regularization parameter.

By using the principle of 1-D variational assimilation, the final iterative formula of the atmospheric profiles is obtained as follows:

$$\mathbf{x}_{n+1} = \mathbf{x}_0 + \left(\mathbf{K}_n^{\mathrm{T}}\mathbf{S}_e^{-1}\mathbf{K}_n + \gamma\mathbf{S}_a^{-1}\right)^{-1} \cdot \mathbf{K}_n^{\mathrm{T}} \cdot \mathbf{S}_e^{-1} \times \left(\mathbf{y}^m - F(\mathbf{x}_n) + \mathbf{K}_n \cdot (\mathbf{x}_n - \mathbf{x}_0)\right) \tag{10}$$

where $\mathbf{x}_n$ represents the result of the nth iteration of the atmospheric profiles, $\mathbf{x}_{n+1}$ represents the result of the $(n+1)$th iteration, $\mathbf{K}_n$ is the Jacobian matrix at $\mathbf{x}_n$, and $F(\mathbf{x}_n)$ represents the simulated radiation spectrum calculated by the radiative transfer model.

IV. Results and Analysis

To verify the accuracy and effectiveness of the IReA constructed in this article, the retrieval results of the method were compared and analyzed under clear-sky conditions and cloudy conditions. Under clear-sky conditions, the discussion focuses on the extent to which the accuracy of the inversion of temperature and humidity profiles can be improved by combining MWRs. The advantages of CNNs over traditional neural network models (BP neural networks) and physical retrieval methods are analyzed. Under cloudy conditions, the focus is to compare whether the joint use of temperature and humidity channels can improve the accuracy of retrieval results after combined microwave observation.

A. Analysis Under Clear-Sky Conditions

To investigate the accuracy and effectiveness of the IReA (which is established by combining a ground-based infrared hyperspectral radiometer and a MWR), the retrieved atmospheric temperature and humidity profiles of the IReA, BP neural network, and physical retrieval methods were obtained. The physical retrieval method is the classical AERI retrieval algorithm, and a comparison of the IReA and physical retrieval methods is performed to investigate whether the retrieval accuracy can be improved by combining the MWR observations under clear-sky conditions. Currently, the BP method is also a commonly used machine learning method for retrieving atmospheric profiles. Therefore, to investigate the advantages of CNN over traditional neural networks, we further compare the results of the IReA with those of the BP neural network.

1) Case Analysis Under Clear-Sky Conditions: To investigate the performance of the IReA, the retrieval process is first performed under the condition of a "temperature inversion layer." In this case, the atmospheric temperature above the ground abnormally increases with height, and the variation in the temperature profile is more complex than that in the normal profiles. Therefore, the accuracy and robustness of the inversion model can be effectively tested by its ability to capture special atmospheric structures in inverted profiles.

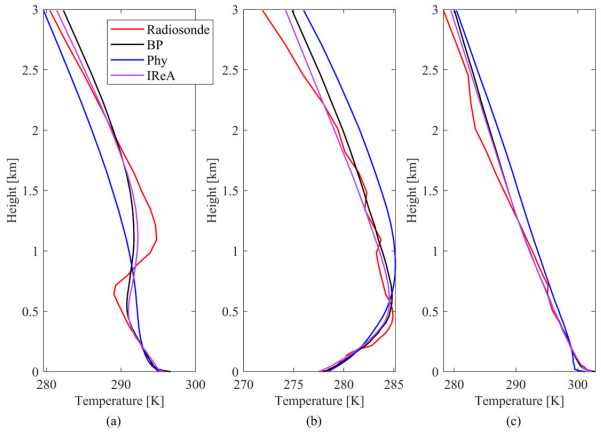


Fig. 4. Temperature profiles were retrieved by the BP algorithm, physical algorithm, and IReA. The red line is the radiosonde value, the black line indicates the result of the BP method, the blue line indicates the result of the physical retrieval method, and the purple line indicates the result of the IReA method. (a) April 8th, (b) April 11th, and (c) June 1st, in 2017.

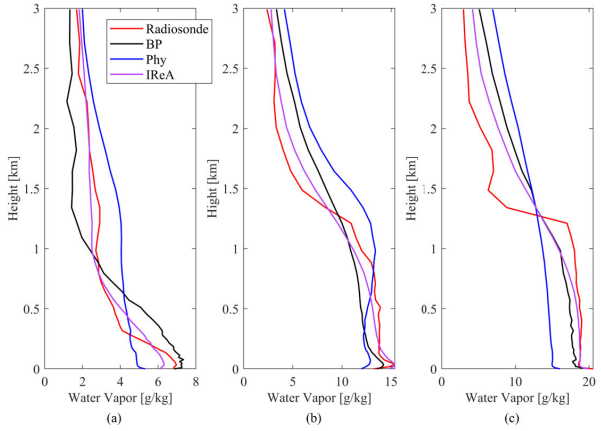


Fig. 5. Humidity profiles were retrieved by the BP algorithm, physical algorithm, and IReA. Same as Fig. 4 but for humidity profiles. (a) April 8th, (b) April 11th, and (c) June 1st, in 2017.

In this test, three cases are selected for validation, that is, the observations on April 8th, April 11th, and June 1st in 2017. On April 8th, a temperature inversion occurred at an altitude of 0.5–1 km, and on April 11th, there was a temperature inversion from the ground to an altitude of 0.5 km. For the temperature profiles on June 1st, a weaker "temperature inversion" phenomenon is observed at a height of 2 km. The temperature and humidity profiles retrieved by the IReA are compared with those of the BP neural network and physical retrieval method, and the results are shown in Figs. 4 and 5, respectively.

Fig. 4 shows a comparison of the temperature profiles obtained by the radiosonde, BP, physical algorithm, and IReA methods. Although the retrieval profiles of all three methods show a good agreement with the radiosonde data at 500 m, the retrieval accuracies of the IReA and BP methods are slightly greater than that of the physical retrieval algorithm. For an atmosphere above 500 m, their abilities to capture the variation characteristics of the stratification of the atmosphere are different, especially for transitions of the temperature

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curves. From Fig. 5, it can also be found that the retrieved curves of the physical retrieval method pass through the curves of the true value in a smoother way, while the IReA and the BP methods have a better response for the temperature transitions. Compared with the results retrieved by the BP method, the IReA retrieval curves are closer to the actual radiosonde data. The phenomena above can be explained by the following reasons. In the physical algorithm, only the infrared radiance information of AERI is used, while for the IReA and BP networks, a combination of infrared and microwave observations is adopted, therefore the information content of the IReA and BP networks is richer than that of the physical algorithm. Compared with the BP method, the IReA network has a stronger ability to extract features of different scales and different levels; therefore, its retrieval accuracy is greater than that of the BP neural network.

Fig. 5 displays a comparison of the retrieved humidity profiles with the corresponding radiosonde data. Overall, the retrieval results of the IReA and BP methods are closer to the curves of the radiosonde data than the physical algorithm, and the IReA exhibits a more robust ability to capture the turning points of the humidity profiles. This conclusion is similar to that in the retrieval of the temperature, which indicates that adding microwave observations can effectively improve the information content of the water vapor. Fig. 5 also shows that the retrieval accuracy of the IReA is greater than that of the BP method, especially in the atmosphere above 1 km, where the maximum retrieval error of the IReA in Fig. 4(b) is 1.7 g/kg, while for the BP method, the maximum error is 2.8 g/kg. From the analysis above, it can be seen that the nonlinear fitting ability of IReA is better than that of the BP method due to the special structure of the network.

On the whole, the inversion results of the three methods have deviations from the results of radiosondes, especially in the upper atmosphere. The reasons include two aspects: on the one hand, the weight function of ground-based infrared hyperspectral radiation is mainly concentrated in the lower atmosphere, and in the upper atmosphere, the amount of information in hyperspectral radiation is relatively small. On the other hand, the radiosonde and AERI themselves are subject to instrument observation errors. For the radiosonde, the temperature error is about ± 0.2 °C, and the humidity error is about $\pm 2\%$. For AERI, though after the calibration of the responsivity and offset, there is still some error in the measurement, about 1 Kelvin at the known temperatures (290 and 330 K). Besides, the detection process of the radiosonde requires a long time, during which the atmospheric state may change to some extent. However, our inversion experiment matches observation data based on the release time of the radiosonde, and the time for comparing the upper atmosphere is not consistent.

2) Statistical Analysis Under Clear-Sky Conditions: To further validate the performance of IReA, the temperature and humidity profiles in 2017 (including 802 samples) were inverted, and the RMSE and MAE were calculated for the BP method, physical retrieval algorithm, and IReA. The results are shown in Table II. The MAE and RMSE of the physical retrieval algorithm are 2.7068 and 1.5835 K. The variation of the BIAS of the retrieval results with height for the physical retrieval method is also consistent with Turner's method [43]. The BP method has a relatively better performance in the retrieval results of the temperature profile, and the MAE and RMSE are 0.8921 and 0.8723, respectively, which indicates that the retrieval result is closer to the true value.

TABLE II
AVERAGE RETRIEVAL ERRORS OF THE BP METHOD, PHYSICAL RETRIEVAL METHOD, AND IREA

Temperature (K)	MAE	RMSE
BP	0.8921	0.8723
Phy	2.7068	1.5835
IReA	0.7343	0.7969
Water Vapor (g/kg)	**MAE**	**RMSE**
BP	1.1517	1.0476
Phy	2.1655	1.4426
IReA	0.9715	0.9523

For the temperature profiles, compared with the BP method, the IReA demonstrated an improvement of 18% in terms of accuracy, and the RMSE and MAE of the IReA were 0.7969 and 0.7343 K, respectively. In addition, an improvement of 73% in the MAE can be achieved in comparison with that of the physical retrieval method. In terms of the RMSE, the IReA demonstrated an improvement of 9% compared with that of the BP method and an improvement of nearly 50% compared with that of the physical retrieval method.

For the retrieval of humidity profiles, the BP method has relatively low MAE and RMSE on the retrieval results, indicating that the deviation between its results and the true values is small. However, the accuracy decreases when compared to the retrieval results of temperature profiles. The retrieval results of the physical retrieval method have relatively high MAE and RMSE, indicating a large deviation of the results from the true values. Similar to the results for temperature, the accuracy of the Phy method on the retrieval results of humidity profiles is not as good as that of the BP and IReA methods. The RMSE and MAE of the IReA are 0.9715 and 0.9523 g/kg, respectively, demonstrating an improvement of 16% compared with those of the BP method. According to the statistical results, the performance of the IReA is also better than that of the physical retrieval method, and its retrieval result of the MAE is improved by 55%. In terms of the RMSE, the IReA indicates an improvement of 9% compared with that of the BP algorithm and an improvement of 34% compared with that of the physical retrieval method. From the statistical analysis, it can be concluded that the RMSE and MAE values obtained by IReA are the smallest among the three methods, followed by those obtained by the BP method and the physical retrieval method.

Fig. 6 illustrates the retrieval accuracy (BIAS and RMSE) for temperature and humidity profiles using the BP method, physical method, and IReA, based on 802 profiles from the validation set under clear-sky conditions. Overall, the BIAS of the IReA is smaller than that of the physical retrieval method and the BP method, especially at heights between 500 and 2500 m. The BIAS of the physical method is the

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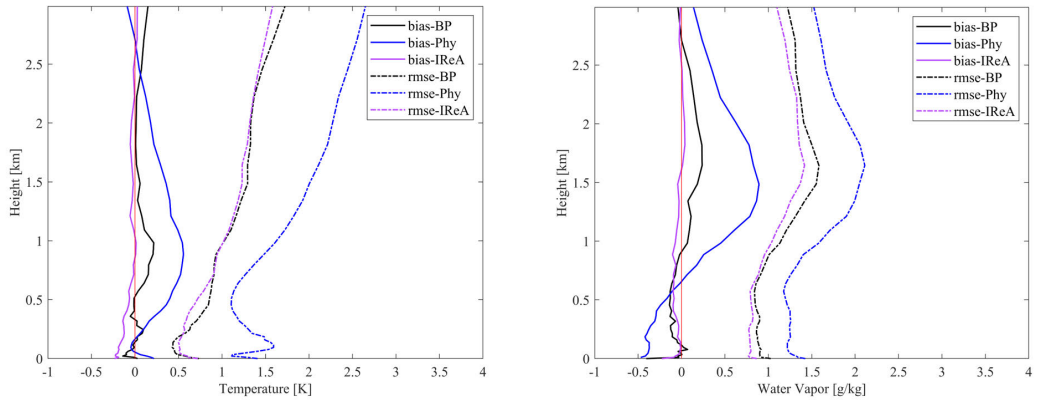


Fig. 6. Retrieval accuracy (BIAS and RMSE) of temperature and humidity profiles by the BP method, physical retrieval method, and IReA. The left figure shows the temperature results and the right figure shows the water vapor results. The black line represents the BP method results, the blue line represents the physical retrieval method results, and the purple line represents the IReA results. The colored solid lines represent the BIAS of each method, while the colored dashed lines represent the RMSE. The vertical red line represents zero values.

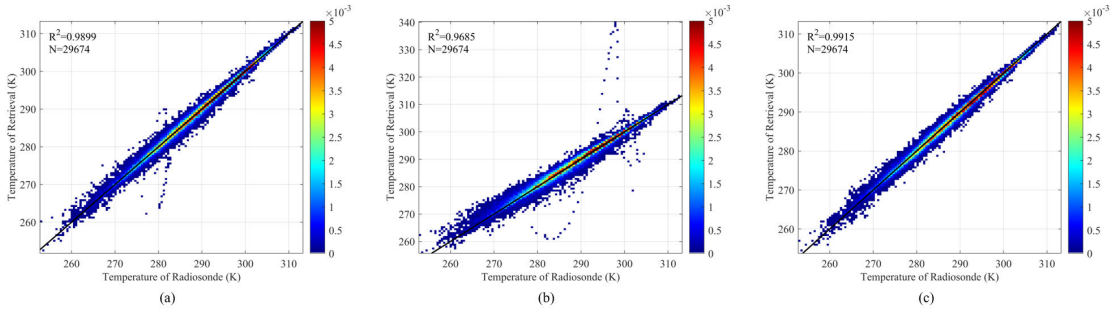


Fig. 7. Scatter comparison between temperature profiles and radiosonde data retrieved by (a) BP algorithm, (b) physical retrieval algorithm, and (c) IReA.

largest, approximately 0.5 K at a height of approximately 1000 m. According to the vertical distribution of the BIAS, the retrieval temperature is generally greater than the temperature obtained by radiosondes, while for the IReA, its retrieval value is slightly smaller than the true value. The RMSEs of all these models increase with increasing height, where the RMSE of the IReA is slightly smaller than that of the BP method, while the RMSE of the physical method is notably larger than those of the IReA and BP methods.

Similar phenomena can be observed for the water vapor profile. Fig. 6 shows that the IReA exhibits a negative BIAS, whereas the BP algorithm and physical retrieval method exhibit a predominantly positive BIAS. The results indicate that the humidity profile retrieved by the IReA is generally larger than the true value, while the opposite conclusion is obtained for the BP algorithm and physical retrieval method. Although the BIAS values of the three retrieval algorithms are not significantly different in the lower atmosphere, the BIAS values of the BP method and the physical retrieval method gradually increase when the altitude is greater than 500 m and reach the maximum value at a height of 1 km. The performance of the IReA is superior to that of the BP method and physical retrieval method, and its BIAS is generally less than 0.1 g/kg. For the RMSE of the water vapor profile, the RMSE increases with height overall under 1500 m and then decreases at altitudes above 1500 m. The accuracy of the IReA is obviously the highest among the three models, followed by the BP method, where the RMSE of the IReA is generally within 1.4 K.

To further investigate the performances of IReA, physical retrieval method, and BP method, scatterplots of their retrieval results and radiosonde data are drawn with a total of 29 674 data points, where Fig. 7 shows the correlation between the retrieved temperature and radiosonde data, while Fig. 8 shows a comparison of the humidity profiles obtained by the different models. Overall, the scatter plots of the temperature profiles and humidity profiles are consistent with those of the RMSE and MAE, showing that the IReA has the highest correlation coefficient R^2 with the radiosonde data, with a correlation coefficient of 0.9915 for temperature and 0.9517 for humidity. The R^2 of the temperature profile retrieved by the IReA is 0.0016 greater than that retrieved by the BP method and 0.023 greater than that retrieved by the physical retrieval method. For the humidity profiles, the R^2 of the IReA method is 0.0097 greater than that of the BP method and 0.0569 greater than that of the physical retrieval method. From the analysis above, it can be found that the IReA shows the best nonlinear fitting ability, and the retrieval accuracy of the BP algorithm and physical retrieval method is slightly lower than that of the IReA overall. Comparing the scatter plots of the temperature and the humidity profiles, the performance of the three methods for water vapor is worse than that for temperature, which is similar to the results in Section IV-A2. This phenomenon might be explained by the irregular vertical distribution of

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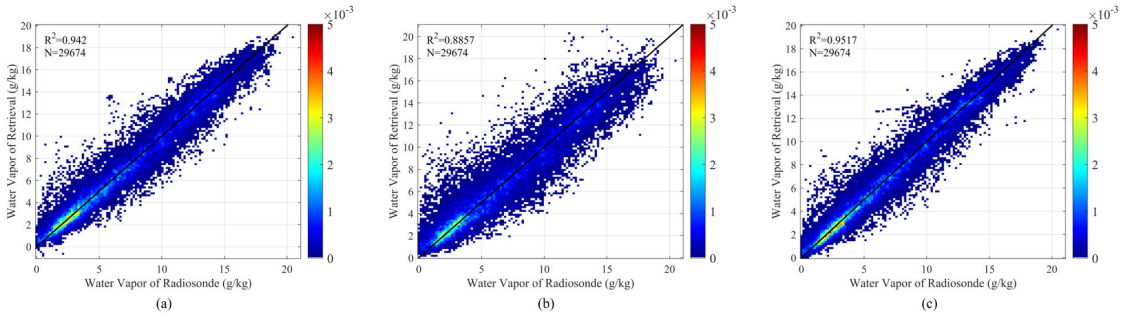


Fig. 8. Scatter comparison between humidity profiles and radiosonde data retrieved by (a) BP algorithm, (b) physical retrieval algorithm, and (c) IReA.

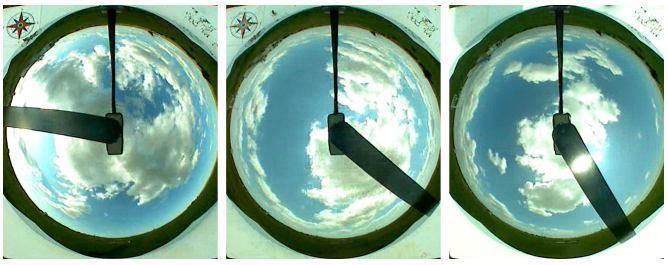


Fig. 9. TSI cloud images corresponding to individual cases (three-time points: 23:30, 17:30, and 17:45).

water vapor in the atmosphere, which is much more complex than temperature.

B. Analysis Under Cloud Conditions

In the previous part of the study, we only explored the performance of the IReA under clear-sky conditions. However, the retrieval accuracy under cloudy conditions has not yet been discussed. Therefore, to gain a deeper understanding of the model's performance under cloudy conditions, 1110 sets of data under cloudy conditions were selected to assess the effectiveness of the IReA. In this test, the temperature and humidity are first retrieved by the joint retrieval model ($IReA_1$), which uses both infrared hyperspectral and microwave observations, and the results are compared with the results that use only infrared hyperspectral radiance data ($IReA_2$). Similar to Section IV-A, the analysis under cloudy conditions is also divided into two parts: case analysis and statistical analysis.

1) Case Analysis Under Cloud Conditions: In this section, three cloudy cases are selected and analyzed, and the time distributions of these cases are May 30th, June 29th, and August 16th in 2017. Fig. 9 shows the cloud distribution obtained by TSI at the three time points, with thin clouds in the visual mirror.

Figs. 10 and 11 show the comparison results of the temperature and humidity profiles retrieved by $IReA_1$ and $IReA_2$ respectively. Regardless of the temperature or humidity profile, the inversion result of $IReA_1$ is closer to the radiosonde data than that of $IReA_2$, and the improvement in the humidity is much more remarkable than that in the temperature profile. In addition, as shown in Fig. 9, as the cloud cover decreases from left to right, the inversion accuracy of the humidity

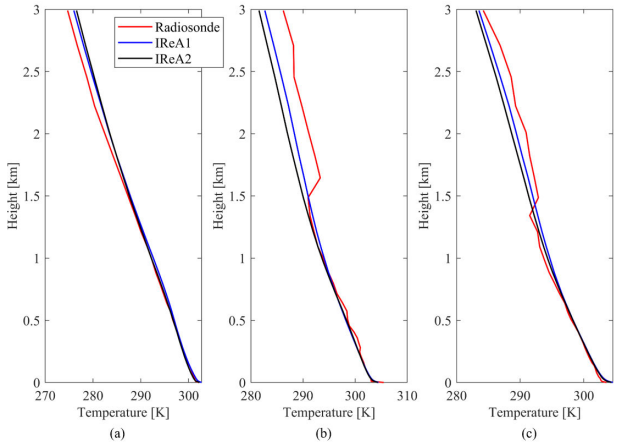


Fig. 10. Comparison between the $IReA_1$ and $IReA_2$ retrieval temperature profiles and radiosonde data. (a) May 30th, (b) June 29th, and (c) August 16th, in 2017

profile also increases. Overall, especially for the changes in temperature and humidity with height, $IReA_1$ is more consistent with the radiosonde data than $IReA_2$. The results indicate that the addition of microwave radiance observations can effectively reduce the influence of clouds on the retrieval of temperature and humidity so that the retrieval accuracy is improved to some extent, especially for humidity profiles. Although the improvement is relatively limited, we believe that if more microwave channels are introduced, the retrieval accuracy of the temperature and humidity profiles of the IReA will improve more notably.

2) Statistical Analysis Under Cloud Conditions: Similar to the results for clear-sky conditions, the RMSE and MAE of temperature and humidity were calculated for the retrieval results of $IReA_1$ and $IReA_2$, and the results are presented in Table III, where the radiosonde data are taken as the true values. The RMSE and the MAE obtained by $IReA_1$ are smaller than those obtained by $IReA_2$, which indicates that $IReA_1$ has a better performance in the retrieval of temperature and humidity. For the retrieval of temperature profiles, the RMSE of $IReA_1$ is 1.1398 K and the MAE is 1.6053 K, which is an improvement of 1% compared to that of $IReA_2$. In terms of the RMSE, $IReA_1$ presented an improvement of 4% compared to $IReA_2$. For the retrieval of the humidity profiles, $IReA_1$ has an RMSE of 0.9749 K and an MAE

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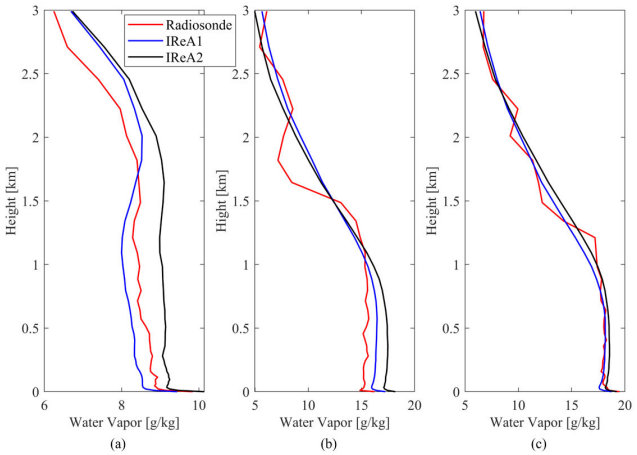


Fig. 11. Comparison between the $IReA_1$ and $IReA_2$ retrieval humidity profiles and radiosonde data. (a) May 30th, (b) June 29th, and (c) August 16th, in 2017.

TABLE III
AVERAGE RETRIEVAL ERRORS OF THE IREA (1 MEANS COMBINED WITH INFRARED AND MICROWAVE CHANNELS, 2 MEANS WITHOUT COMBINATION)

Temperature (K)	RMSE	MAE
$IReA_1$	1.1398	1.6053
$IReA_2$	1.1536	1.6654
Water Vapor (g/kg)	**MAE**	**RMSE**
$IReA_1$	0.9749	1.0070
$IReA_2$	1.0307	1.1277

of 1.0070 g/kg, which reflects a 5% increase in the RMSE compared with that of $IReA_2$ and an 11% increase in the MAE compared to that of $IReA_2$. Overall, the improvement in the humidity profiles of $IReA_1$ is more remarkable than that in the temperature profiles. This suggests that adding microwave observations might effectively improve the inversion accuracy of atmospheric profiles under cloudy conditions.

Fig. 12 further shows the vertical distributions of the MAEs and RMSEs of $IReA_1$ and $IReA_2$ for the retrieved temperature and humidity profiles, utilizing 1110 profiles from the validation set under cloudy conditions. Where the temperature results are on the left and the water vapor data are on the right. For the temperature profiles, $IReA_1$ begins to outperform $IReA_2$ at an altitude of 1000 m, and both the MAE and RMSE are smaller than those of $IReA_2$. For the water vapor profiles, $IReA_1$ outperforms $IReA_2$ within all the inversion height layers. This analysis further validates the conclusions drawn in Table III.

Similar to Section IV-A2, scatter plots between the retrieval results and radiosonde data are drawn to investigate the performances of $IReA_1$ and $IReA_2$ under cloudy conditions, see Figs. 13 and 14. The scatter plots of the temperature and humidity profiles are consistent with the RMSE and MAE results, and the correlation coefficients between the $IReA_1$ and radiosonde data are greater than those of $IReA_2$. The correlation coefficient of temperature and humidity between $IReA_1$ and radiosondes is 0.9822, and that of humidity is 0.9484, which is 0.4% greater than that of $IReA_2$, indicating that the retrieval performance of IReA is better under cloudy conditions if microwave observations are combined with infrared radiance data. Overall, the improvement in the humidity profile is greater than that in the temperature profile, indicating that the inversion accuracy of humidity under cloudy conditions is improved more notably than that of temperature after the addition of microwave data.

TABLE IV
TIME CONSUMPTION OF THREE RETRIEVAL METHODS

Time	Methods		
	Phy	BP	IReA
02/15	19.75 min	0.4 s	3.1 s
03/18	18.68 min	0.2 s	2.9 s
04/11	19.73 min	0.3 s	3.4 s
05/09	18.73 min	0.2 s	3.1 s
07/10	18.89 min	0.2 s	3.0 s
08/24	19.37 min	0.3 s	3.2 s
10/25	19.29 min	0.3 s	3.3 s
11/21	19.06 min	0.2 s	3.1 s
12/16	19.19 min	0.3 s	3.3 s

V. DISCUSSION

In order to verify the effectiveness of the method, we further compared the time consumption of these methods as shown in Table IV. Currently, the speed of training CNNs in IReA is 6 min, and the speed of applying the trained CNNs to retrieve a profile is about 3 s. The speed of training the BP neural network is about 2 min, and the speed of applying the trained BP to retrieve a profile is about 0.2 s. The computational efficiency of the physical retrieval method is assessed based on the time required to achieve convergence in output profiles after seven iterations. The computational efficiency of the BP and IReA methods is significantly higher than the physical retrieval method. This superiority is attributable to the intricate physical processes inherent in the physical retrieval approach. Although our proposed method exhibits a longer training duration compared to the BP method, it remains within acceptable bounds.

CNNs are susceptible to overfitting when processing small datasets, primarily because they typically necessitate extensive training data to effectively learn intricate features. Furthermore, the decision-making process within CNNs is often perceived as a "black box," hindering interpretability in real-world applications due to the complex interplay of multilayer convolution and pooling operations. Overfitting predominantly arises from the high model complexity inherent to CNNs, which when trained on limited data, risks memorizing noise rather than generalizing to unseen instances. In this study, we computed the model parameters, and the number of training sets in this article is sufficient to fit these parameters.

In this article, only microwave and infrared data are used for the retrieval of atmospheric profiles. Multiinstrument and multidata studies based on ground-based infrared hyperspectral data deserve further exploration to obtain more accurate and comprehensive results. In addition, due to the limitation of the MWR of SGP, only two channels are used in our retrieval algorithm, if more channels are used, the retrieval accuracy might improve further, and meaningful work can be done in the future. For example, we can utilize multichannel MWR

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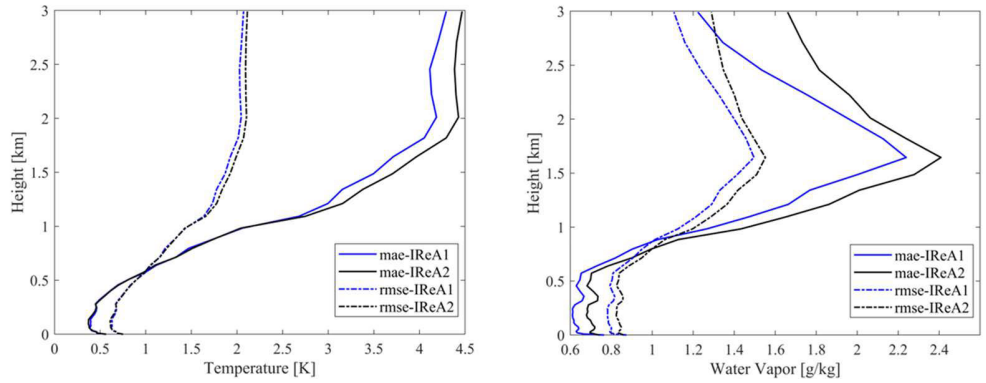


Fig. 12. Retrieval accuracy (MAE and RMSE) of the temperature and humidity profiles of the IReA (1 means combined infrared and microwave channels and 2 means not combined). The blue line represents the $IReA_1$ results and the black line represents the $IReA_2$ results. The solid line represents the MAE and the dashed line represents the RMSE.

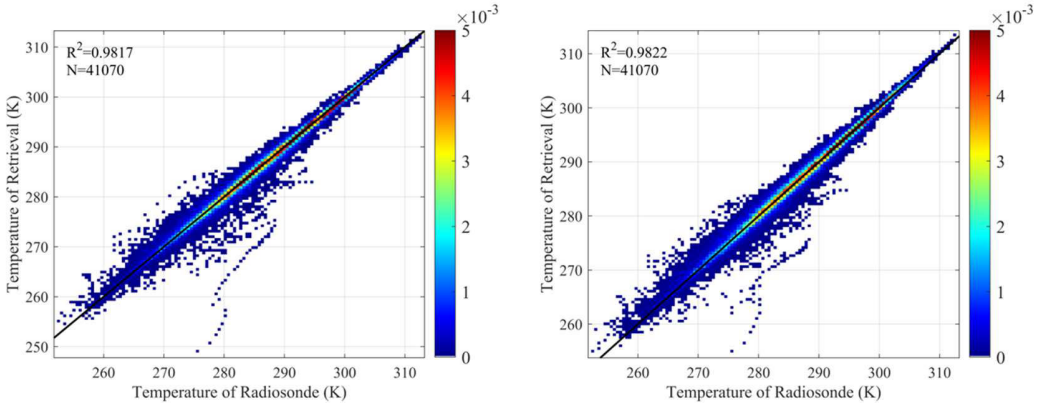


Fig. 13. Scatter comparison between temperature profiles retrieved from $IReA_1$ and $IReA_2$ and radiosonde data.

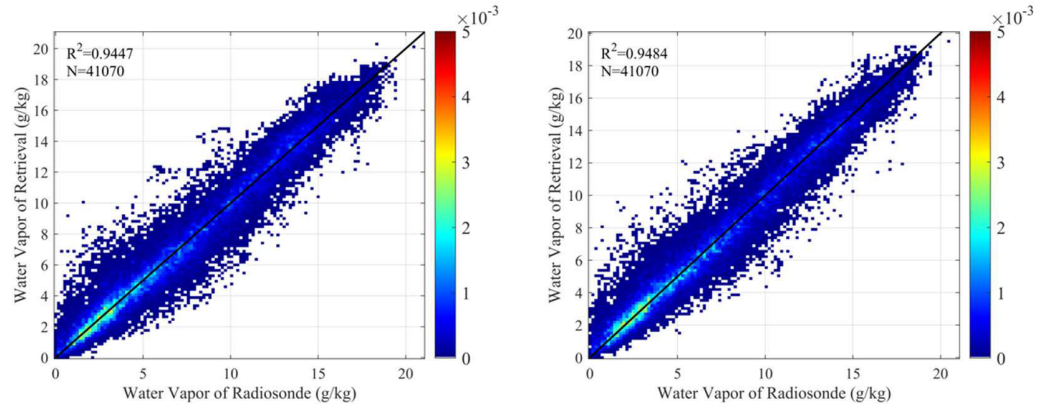


Fig. 14. Scatter comparison between humidity profiles retrieved from $IReA_1$ and $IReA_2$ and radiosonde data.

data and AERI data to improve the quality and diversity of training data. Second, model optimization techniques such as hyperparameter tuning are used to reduce the risk of overfitting and improve the generalization ability of the model. It can also be combined with physical retrieval methods to increase the physical process of the model. Our method relies on the completeness of the training dataset, and machine-learning models are often "black box" models that have difficulty explaining their internal decision-making processes. While our method exhibits superiority over physical retrieval methods, it lacks physical interpretability, it is an important work we will study in our next step, and we are now fitting only for SGP sites, the model accuracy depends on the completeness of the data, the next step will be to collect more data to improve the generalization of the model.

VI. Summary

To improve the retrieval accuracy of the atmospheric temperature and humidity profiles of the boundary layer, an IReA

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based on a CNN is proposed by combining ground-based infrared hyperspectral observation data and MWR data. To validate the effectiveness of the IReA model, the performances of the BP algorithm, the physical retrieval method, and the IReA were systematically compared and analyzed under clear-sky and cloudy conditions.

The main conclusions are drawn as follows.

1) The combination the infrared and microwave observations can effectively improve the retrieval accuracy of atmospheric temperature and humidity profiles under clear-sky conditions. The RMSE of the temperature profile retrieval of the IReA is 0.79 K, and the RMSE of the humidity profile is 0.95 g/kg, which is an improvement of at least 34% compared to that of the physical retrieval algorithm that only uses infrared observations.
2) The IReA has shown a strong nonlinear fitting ability in the retrieval of temperature and humidity profiles. The retrieval accuracy of the IReA is greater than that of the BP network, indicating that the IReA network has a greater ability to extract features of different scales and different levels.
3) Under cloudy conditions, the results show that by combining the microwave channel as a complement, the retrieval accuracy of the IReA can be improved, especially for humidity. The RMSE and MAE of the water vapor data obtained by IReA_1 are 1.0070 and 0.9749 g/kg, respectively, which are lower than those obtained by IReA_2 on the whole. In addition, the correlation coefficient R^2 of IReA_1 is greater than that of IReA_2.

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